

2018 NYJC J2 H2 Mathematics Preliminary exam 9758/2 Marking Guide

Qn	Suggested Answers			
1(i)	$u_1 = a = 4, u_{25} = 1 \Rightarrow ar^{24} = 1$ $\Rightarrow r = \sqrt[24]{0.25} \text{ or } 0.94387$ $S_{25} = \frac{a(1-r^{25})}{1-r} = \frac{4[1-(\sqrt[24]{0.25})^{25}]}{1-\sqrt[24]{0.25}} \approx 54.451 \approx 54.5$ or $S_{25} = \frac{a(1-r^{25})}{1-r} = \frac{4[1-(0.94387)^{25}]}{1-0.94387} \approx 54.449 \approx 54.4$			
1(ii)	$S_{\infty} = \frac{a}{1-r} = \frac{4}{1-\sqrt[24]{0.25}} \approx 71.269 < 71.3$ or $S_{\infty} = \frac{a}{1-r} = \frac{4}{1-0.94387} \approx 71.263 < 71.3$ Hence theoretical maximum total amount of chemical she uses will not exceed 71.3 grams.			
	$u_{25} = 1 \Rightarrow a + 24d = 1 \text{ -----(1)}$ $S_{25} = \frac{25}{2}(2a + 24d) = 54.451$ $\Rightarrow 25a + 300d = 54.451 \text{ -----(2)}$ Solving the 2 equations, $d \approx -0.0982$ and $a \approx 3.36$ or $S_{25} = \frac{25}{2}(2a + 24d) = 54.449 \text{ -----(2)}$ $\Rightarrow 25a + 300d = 54.449$ Solving the 2 equations, $d \approx -0.0982$ and $a \approx 3.36$			

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	or $S_{25} = \frac{25}{2}(2a + 24d) = 54.45 \text{-----} (2)$ $\Rightarrow 25a + 300d = 54.45$ Solving the 2 equations, $d \approx -0.0982$ and $a \approx 3.36$			
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2(a)(i)	$\int \frac{2x-4}{x^2-2x+4} dx = \int \frac{2x-2}{x^2-2x+4} dx - \int \frac{2}{x^2-2x+4} dx$ $= \int \frac{2x-2}{x^2-2x+4} dx - \int \frac{2}{(x-1)^2+3} dx$ $= \ln(x^2-2x+4) - \frac{2}{\sqrt{3}} \tan^{-1} \frac{x-1}{\sqrt{3}} + C$			
2(a)(ii)	$x^2-2x+4 = (x-1)^2+3 > 0 \text{ since } (x-1)^2 \geq 0.$ Therefore, $\int_1^4 \frac{ 2x-4 }{x^2-2x+4} dx = -\int_1^2 \frac{2x-4}{x^2-2x+4} dx + \int_2^4 \frac{2x-4}{x^2-2x+4} dx$ $= -\left[\ln(x^2-2x+4) - \frac{2}{\sqrt{3}} \tan^{-1} \frac{x-1}{\sqrt{3}} \right]_1^2 + \left[\ln(x^2-2x+4) - \frac{2}{\sqrt{3}} \tan^{-1} \frac{x-1}{\sqrt{3}} \right]_2^4$ $= -\left[\left(\ln(4) - \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} \right) - \left(\ln(3) - \frac{2}{\sqrt{3}} \tan^{-1} 0 \right) \right] +$ $\left[\left(\ln(12) - \frac{2}{\sqrt{3}} \tan^{-1} \frac{3}{\sqrt{3}} \right) - \left(\ln(4) - \frac{2}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} \right) \right]$ $= -2\ln 4 + \ln 3 + \ln 12 + \frac{4}{\sqrt{3}} \left(\frac{\pi}{6} \right) - \frac{2}{\sqrt{3}} \left(\frac{\pi}{3} \right)$ $= \ln \left(\frac{9}{4} \right) = 2 \ln \left(\frac{3}{2} \right)$			
2(b)				

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$$\int_0^{\frac{\pi}{4}} \frac{\sin^2 x}{\cos^3 x} dx = \frac{1}{2} \int_0^{\frac{\pi}{4}} \frac{2 \sin x}{\cos^3 x} (\sin x) dx$$

$$= \frac{1}{2} \left[\left[\frac{\sin x}{\cos^2 x} \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \frac{1}{\cos x} dx \right]$$

$$= \frac{1}{2} \left[\left[\frac{\sin x}{\cos^2 x} \right]_0^{\frac{\pi}{4}} - \int_0^{\frac{\pi}{4}} \sec x dx \right]$$

$$= \frac{1}{2} \left[\frac{\sin x}{\cos^2 x} - \ln(\sec x + \tan x) \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{\sqrt{2}} - \frac{1}{2} \ln(\sqrt{2} + 1)$$

$$u = \sin x, \frac{dv}{dx} = \frac{2 \sin x}{\cos^3 x}$$

$$\frac{du}{dx} = \cos x, v = \frac{1}{\cos^2 x}$$

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Qn	Suggested Answers			
3(i)	$z^3 + 4z^2 + 8z + 8 = (z + 2)(z^2 + az + 4)$ Comparing coefficient of z^2: $4 = a + 2 \Rightarrow a = 2$ $\therefore z^2 + 2z + 4$ is a factor of $z^3 + 4z^2 + 8z + 8$ $(z + 2)(z^2 + 2z + 4) = 0$ $z + 2 = 0$ or $z^2 + 2z + 4 = 0$ $z = -2$ or $z = \frac{-2 \pm \sqrt{4 - 16}}{2} = -1 \pm i\sqrt{3}$ $\therefore z_1 = -1 + i\sqrt{3}$ and $z_2 = -1 - i\sqrt{3}$			
3(ii)	$ w = \frac{ z_1 }{ z_2 } = 1$ $\arg w = \arg \left(\frac{z_1}{z_2} \right) = \arg z_1 - \arg z_2$ $= \frac{2\pi}{3} - \left(-\frac{2\pi}{3} \right) = \frac{4\pi}{3}$ $\therefore \arg w = \frac{4\pi}{3} - 2\pi = -\frac{2\pi}{3}$			
3(iii)	$w^n = e^{-i\frac{2n\pi}{3}} = \cos\left(-\frac{2n\pi}{3}\right) + i\sin\left(-\frac{2n\pi}{3}\right)$ Since w^n is real, $\sin\left(-\frac{2n\pi}{3}\right) = 0$ $\sin\left(\frac{2n\pi}{3}\right) = 0$ $\frac{2n\pi}{3} = 0, \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \pm5\pi, \dots$ Since $n \in \mathbb{C}^+$, $n = 3, 6, 9, \dots$			

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	Set of positive integers n is $\{n : n = 3p, p \in \mathbb{C}^+\}$ Or Since w^n is real, $\arg w^n = 0, \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \pm5\pi, \dots$ $-\frac{2n\pi}{3} = 0, \pm\pi, \pm2\pi, \pm3\pi, \pm4\pi, \pm5\pi, \dots$ Since $n \in \mathbb{C}^+, n = 3, 6, 9, \dots$ Set of positive integers n is $\{n : n = 3p, p \in \mathbb{C}^+\}$ $w^n = e^{-i\frac{2n\pi}{3}} = e^{-i2p\pi} = \cos(-2p\pi) + i\sin(-2p\pi) = 1 \text{ (shown)}$			
3(iv)	$w^n = e^{-i\frac{2n\pi}{3}} = \cos\left(-\frac{2n\pi}{3}\right) + i\sin\left(-\frac{2n\pi}{3}\right)$ $w^{100} - (w^*)^{100} = w^{100} - (w^{100})^*$ $= 2i\sin\left(-\frac{200\pi}{3}\right)$ $= 2i\left(-\frac{\sqrt{3}}{2}\right) = -\sqrt{3}i$ Alternatively, $w^{100} - (w^*)^{100} = w^{100} - (w^{100})^*$ $= w - w^* = e^{-i\frac{2\pi}{3}} - e^{i\frac{2\pi}{3}}$ $= -2i\sin\left(\frac{2\pi}{3}\right) = -2i\left(\frac{\sqrt{3}}{2}\right) = -\sqrt{3}i$			

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Qn	Suggested Answers			
4(a)(i)	$y = \ln(x + \sqrt{x^2 + 1})$ $e^y = x + \sqrt{x^2 + 1}$ Differentiating both sides w.r.t. x $e^y \frac{dy}{dx} = 1 + \frac{1}{2}(x^2 + 1)^{-\frac{1}{2}}(2x)$ $\Rightarrow e^y \frac{dy}{dx} = 1 + x(x^2 + 1)^{-\frac{1}{2}}$ Differentiating both sides w.r.t. x $e^y \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx}\right)^2 = (x^2 + 1)^{-\frac{1}{2}} + x\left(-\frac{1}{2}\right)(x^2 + 1)^{-\frac{3}{2}}(2x)$ $\Rightarrow e^y \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx}\right)^2 = (x^2 + 1)^{-\frac{1}{2}} - x^2(x^2 + 1)^{-\frac{3}{2}} \text{ (shown)}$			
4(a)(ii)	$e^y \frac{d^3y}{dx^3} + e^y \frac{d^2y}{dx^2} \left(\frac{dy}{dx}\right) + e^y \left(\frac{dy}{dx}\right)^3 + e^y 2 \left(\frac{dy}{dx}\right) \left(\frac{d^2y}{dx^2}\right)$ $= -\frac{1}{2}(x^2 + 1)^{-\frac{3}{2}}(2x) - 2x(x^2 + 1)^{-\frac{3}{2}} - x^2\left(-\frac{3}{2}\right)(x^2 + 1)^{-\frac{5}{2}}(2x)$ $\Rightarrow e^y \frac{d^3y}{dx^3} + 3e^y \frac{d^2y}{dx^2} \left(\frac{dy}{dx}\right) + e^y \left(\frac{dy}{dx}\right)^3$ $= -x(x^2 + 1)^{-\frac{3}{2}} - 2x(x^2 + 1)^{-\frac{3}{2}} + 3x^3(x^2 + 1)^{-\frac{5}{2}}$ When $x = 0$, $y = 0, \frac{dy}{dx} = 1, \frac{d^2y}{dx^2} = 0, \frac{d^3y}{dx^3} = -1$ $\therefore \sinh^{-1} x = x - \frac{x^3}{6} + \dots$			

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4(b)(i)	$\frac{AB}{AD} = \frac{AC \tan \frac{\pi}{6}}{AC \tan \left(\frac{\pi}{6} + \theta \right)} = \frac{1}{\sqrt{3} \tan \left(\frac{\pi}{6} + \theta \right)} \text{ (shown)}$			
4(b)(ii)	$\begin{aligned} \frac{AB}{AD} &= \frac{1}{\sqrt{3} \tan \left(\frac{\pi}{6} + \theta \right)} = \frac{1 - \tan \frac{\pi}{6} \tan \theta}{\sqrt{3} \left(\tan \frac{\pi}{6} + \tan \theta \right)} \\ &= \frac{1}{\sqrt{3}} \left(\frac{1 - \frac{\tan \theta}{\sqrt{3}}}{\frac{1}{\sqrt{3}} + \tan \theta} \right) = \frac{1}{\sqrt{3}} \left(1 - \frac{\tan \theta}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{3}} + \tan \theta \right)^{-1} \\ &= \left(1 - \frac{\tan \theta}{\sqrt{3}} \right) (1 + \sqrt{3} \tan \theta)^{-1} \approx \left(1 - \frac{\theta}{\sqrt{3}} \right) (1 + \sqrt{3} \theta)^{-1} \\ &= \left(1 - \frac{\theta}{\sqrt{3}} \right) \left(1 + (-1) \sqrt{3} \theta + \frac{(-1)(-2)}{2} (\sqrt{3} \theta)^2 + \dots \right) \\ &= \left(1 - \frac{\theta}{\sqrt{3}} \right) (1 - \sqrt{3} \theta + 3 \theta^2 + \dots) \\ &= 1 - \sqrt{3} \theta + 3 \theta^2 - \frac{\theta}{\sqrt{3}} + \theta^2 + \dots \approx 1 + \left(-\frac{4\sqrt{3}}{3} \right) \theta + 4 \theta^2 \\ a &= -\frac{4\sqrt{3}}{3}, b = 4 \end{aligned}$			

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5(i)	No. of arrangements $= \frac{10!}{2!2!4!} - 1$ $= 37799$			
5(ii)	No. of arrangements =Total no. of ways with all E's together – No. of ways all E's together and C's and L's together – No. of ways all E's together and C's but L's separated – No. of ways all E's together and L's but C's separated $= \frac{(7-1)!}{2!2!} - (5-1)! - (4-1)!^4 C_2 - (4-1)!^4 C_2$ $= 180 - 24 - 36 - 36 = 84$			

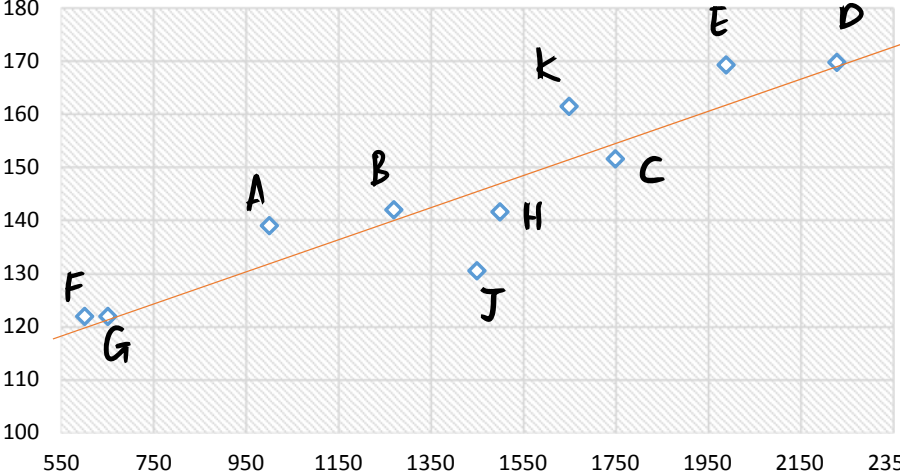
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Qn	Suggested Answers			
6(i)	Let D_i be the random variable denoting the score of the ith throw of the die for $i = 1, 2$ $P(X = 2)$ $= P(D_1 = 1, D_2 = 5) + P(D_1 = 2, D_2 = 4) + P(D_1 = 3, D_2 = 3)$ $+ P(D_1 = 4, D_2 = 2) + P(D_1 = 5, D_2 = 1)$ $= \frac{1}{6} \left(\frac{1}{6} \right) 5 = \frac{5}{36}$			
6(ii)	$P(X > 2 D_1 \text{ is even})$ $= \frac{P(X > 2 \text{ and } D_1 \text{ is even})}{P(D_1 \text{ is even})}$ $= \frac{P(D_1 = 2, D_2 \neq 4) + P(D_1 = 4, D_2 \neq 2)}{P(D_1 \text{ is even})}$ $= \frac{\frac{1}{6} \left(\frac{5}{6} \right) + \frac{1}{6} \left(\frac{5}{6} \right)}{\frac{3}{6}} = \frac{5}{9}$			
6(iii)	Let W be the random variable denoting number of games, out of 10, that a special prize is won. $P(\text{winning a special prize})$ $= P(X = 1) + P(X = 2) = \frac{1}{6} + \frac{5}{36} = \frac{11}{36}$ $W \sim B\left(10, \frac{11}{36}\right)$ Probability required $= P(3 \leq W \leq 8) = P(W \leq 8) - P(W \leq 2)$ $= 0.63173 = 0.632$			

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6(iv)	Let n be the number of games needed. Let Y be the random variable denoting number of games, out of n , that a special prize is won.											
	$Y \sim B\left(n, \frac{11}{36}\right)$											
	$P(Y \geq 1) \geq 0.998 \Rightarrow 1 - P(Y = 0) \geq 0.998$											
	Using GC, <table><tr><td>n</td><td>$1 - P(Y = 0)$</td><td>$1 - P(Y = 0) - 0.998$</td></tr><tr><td>17</td><td>0.99797</td><td>-3×10^{-5}</td></tr><tr><td>18</td><td>0.99859</td><td>5.9×10^{-4}</td></tr></table>	n	$1 - P(Y = 0)$	$1 - P(Y = 0) - 0.998$	17	0.99797	-3×10^{-5}	18	0.99859	5.9×10^{-4}		
n	$1 - P(Y = 0)$	$1 - P(Y = 0) - 0.998$										
17	0.99797	-3×10^{-5}										
18	0.99859	5.9×10^{-4}										
	Least $n = 18$											

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Qn	Suggested Answers			
7(a)	Scatter plot (C) shows perfect negative correlation, so $r_3 = -1$. Scatter plot (A) shows strong, but not perfect, negative correlation, so $-1 < r_1 < 0$. Scatter plot (B) shows no correlation, so $r_2 = 0$. Therefore, $r_3 < r_1 < r_2$.			
7(b)(i)	scatter plot of price (y) against engine capacity (x) 			
7(b)(ii)	Equation of regression line of y on x is $y = 0.03019188x + 102.30964 \approx 0.0302x + 102$			
7(b)(iii)	$r = 0.91336 \approx 0.913$			

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	The value of r implies(suggests) a strong positive linear relationship between x and y .			
	The regression line to be drawn on the scatter diagram passing through D and G. Point corresponding to model J lies well below the line, which implies that its price is lower than it would be expected given its engine capacity. This would be good value for the consumer. Points corresponding to models A, E and K are well above the line, which implies that their prices are higher than it would be expected given their engine capacity and would not be recommended in this case.			

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Qn	Suggested Answers			
8(i)	Let L denotes the mass of a randomly chosen link. $L \sim N(800, 20^2)$ $P(L > 805) \approx 0.40129 \approx 0.401$			
8(ii)	Let S denotes the mass of a randomly chosen link with locking sleeve $S = 1.1L$, $S \sim N(880, 22^2)$ $P(865.35 < S < 895.5) \approx 0.50672 \approx 0.507$			
8(iii)	Let H denotes the mass of a randomly chosen hook $H \sim N(750, \sigma^2)$ $P(H < 735.6) = 0.15$ $\frac{735.6 - 750}{\sigma} = -1.0364$ $\sigma \approx 13.894 \approx 13.9$			
8(iv)	Let C denotes the mass of a wooden box and its contents $C = S_1 + S_2 + S_3 + S_4 + S_5 + H + 1000$ Let $\bar{C} = \frac{C_1 + C_2 + \dots + C_n}{n}$ $\bar{C} \sim N\left(6150, \frac{2613.0}{n}\right)$ $P(\bar{C} > 6190) > 0.013$ $P\left(Z > \frac{6190 - 6150}{\sqrt{\frac{2613.0}{n}}}\right) > 0.013$ $40\sqrt{\frac{n}{2613.0}} < 2.2262$ $n < 8.09$			

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	Greatest $n = 8$											
Qn	Suggested Answers											
9(i)	Let X be the random variable denoting the number of yellow bricks drawn by Donald. P(no yellow bricks) $= P(X = 0)$ $= \frac{1}{2}\left(\frac{5}{10}\right)\frac{1}{2}\left(\frac{4}{9}\right) + \frac{1}{2}\left(\frac{5}{10}\right)\frac{1}{2}\left(\frac{7}{10}\right) + \frac{1}{2}\left(\frac{7}{10}\right)\frac{1}{2}\left(\frac{5}{10}\right) + \frac{1}{2}\left(\frac{7}{10}\right)\frac{1}{2}\left(\frac{6}{9}\right)$ R or B from Box 1 then R or B from Box 1 R or B from Box 1 then R or B from Box 2 R or B from Box 2 then R or B from Box 1 R or B from Box 2 then R or B from Box 2 $= \frac{25}{72}$ (shown)											
9(ii)	$P(X = 2)$ $= \frac{1}{2}\left(\frac{5}{10}\right)\frac{1}{2}\left(\frac{4}{9}\right) + \frac{1}{2}\left(\frac{5}{10}\right)\frac{1}{2}\left(\frac{3}{10}\right) + \frac{1}{2}\left(\frac{3}{10}\right)\frac{1}{2}\left(\frac{5}{10}\right) + \frac{1}{2}\left(\frac{3}{10}\right)\frac{1}{2}\left(\frac{2}{9}\right)$ $= \frac{53}{360}$											
9(iii)	<table border="1" style="margin-left: auto; margin-right: auto;"><tr><td>x</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$P(X = x)$</td><td>$\frac{25}{72}$</td><td>$1 - \frac{25}{72} - \frac{53}{360} = \frac{91}{180}$</td><td>$\frac{53}{360}$</td></tr></table> <i>Direct computation $P(X = 1)$:</i>	x	0	1	2	$P(X = x)$	$\frac{25}{72}$	$1 - \frac{25}{72} - \frac{53}{360} = \frac{91}{180}$	$\frac{53}{360}$			
	x	0	1	2								
$P(X = x)$	$\frac{25}{72}$	$1 - \frac{25}{72} - \frac{53}{360} = \frac{91}{180}$	$\frac{53}{360}$									

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	$P(X=1) = P(Y, Y') + P(Y', Y)$ $= \frac{1}{2} \left(\frac{5}{10} \right) \left[\frac{1}{2} \left(\frac{5}{9} \right) + \frac{1}{2} \left(\frac{7}{10} \right) \right] + \frac{1}{2} \left(\frac{3}{10} \right) \left[\frac{1}{2} \left(\frac{5}{10} \right) + \frac{1}{2} \left(\frac{7}{9} \right) \right]$ $+ \frac{1}{2} \left(\frac{5}{10} \right) \left(\frac{1}{2} \left(\frac{5}{9} \right) + \frac{1}{2} \left(\frac{3}{10} \right) \right) + \frac{1}{2} \left(\frac{7}{10} \right) \left(\frac{1}{2} \left(\frac{5}{10} \right) + \frac{1}{2} \left(\frac{3}{9} \right) \right)$ $= \frac{113}{720} + \frac{23}{240} + \frac{77}{720} + \frac{7}{48} = \frac{91}{180}$ $E(X) = 0 + \frac{91}{180} + 2 \left(\frac{53}{180} \right) = \frac{4}{5}$			
9(iv)	Let W be the random variable denoting the number of yellow bricks drawn out of 5. $W \sim B\left(5, \frac{8}{20}\right)$ P(draws less than 2 yellow bricks) $= P(W < 2) = P(W \leq 1) = 0.33639 = 0.337 \text{ (3 s.f.)}$ Alternatively, P(draws less than 2 yellow bricks) $= \left(\frac{12}{20}\right)^5 + {}^5C_4 \left(\frac{12}{20}\right)^4 \left(\frac{8}{20}\right) = \frac{1053}{3125} \text{ (or 0.33696)}$			
9(v)	P(not more than 9 draws to get the first yellow brick) $= P(\text{first yellow brick is drawn on the 1}^{\text{st}}, 2^{\text{nd}}, \dots \text{or } 9^{\text{th}} \text{ draw})$ $= \left(\frac{8}{20}\right) + \left(\frac{12}{20}\right)^1 \left(\frac{8}{20}\right) + \left(\frac{12}{20}\right)^2 \left(\frac{8}{20}\right) + \dots + \left(\frac{12}{20}\right)^8 \left(\frac{8}{20}\right)$ $= \frac{0.4(1-0.6^9)}{1-0.6}$			

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	$=0.98992 = 0.990$ (to 3 sig. fig.) Alternatively: $P(\text{not more than 9 draws to get the first yellow brick})$ $= 1 - P(10 \text{ or more draws to get the first yellow brick})$ $= 1 - \left(\frac{12}{20}\right)^9 = 0.98992 = 0.990$ (to 3 sig.fig.)			
Qn	Suggested Answers			
10(i)	The sample is random would mean that the result of each throw has the same chance of being selected. The outcome of each throw is also independent of one another.			
10(ii)	Unbiased estimate of the population mean, $\bar{x} = \frac{120}{60} + 65 = 67$ Unbiased estimate of the population variance, $s^2 = \frac{1}{59} \left[\sum (x - 65)^2 - \frac{(\sum (x - 65))^2}{60} \right] = 60.50847 = 60.5$ (to 3 s.f.)			
10(iii)	Let μ be the population mean distance of the throws. Null hypothesis, $H_0 : \mu = 68$ Alt hypothesis, $H_1 : \mu > 68$			
10(iv)	Let X be the random variable denoting the distances thrown by the thrower. Under H_0 , $\bar{X} \sim N\left(68, \frac{7.5^2}{30}\right)$ and test statistic $Z = \frac{\bar{X} - 68}{\frac{7.5}{\sqrt{30}}} \sim N(0,1)$ Critical Region : Reject H_0 if $Z_{\text{calc}} \geq 2.05375$			

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	Calculations : $Z_{\text{calc}} = \frac{\bar{x} - 68}{7.5 / \sqrt{30}}$ Since H_0 is rejected, $\frac{\bar{x} - 68}{7.5 / \sqrt{30}} \geq 2.05375$ $\bar{x} \geq 70.8122$ $\bar{x} \geq 70.9$			
10(v)	If H_0 is not rejected, $Z_{\text{calc}} < 2.05375$ $\frac{70.1 - 68}{7.5 / \sqrt{n}} < 2.05375$ $\sqrt{n} < 7.33482$ $n < 53.79954$ Set of values of n is $\{n \in \mathbb{C}^+ : n \leq 53\}$ or $\{n \in \mathbb{C} : 1 \leq n \leq 53\}$			
	Null hypothesis, $H_0 : \mu = \mu_0$ Alt hypothesis, $H_1 : \mu < \mu_0$ Reject H_0 if $Z_{\text{calc}} \leq -1.64485$ For Ang's conclusion (H_0 is rejected) : $Z_{\text{calc}} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \leq -1.64485$ For Tan's case : $Z'_{\text{calc}} = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{4n}} = 2 \left(\frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} \right)$			

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	Since $\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \leq -1.64485$, $2 \left(\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} \right) \leq 2(-1.64485) \leq -1.64485$, ie $Z'_{\text{calc}} \leq -1.64485$, which leads to H_0 being rejected. Therefore, Tan's test would yield the same conclusion.			
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